
Implementing Neural Style Transfer

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Abstract

In this project, we implemented, evaluated, and modified Neural Style Transfer. Following our implementation of the original model structure, we tested the model's ability to transfer the styles from three widely-known paintings spanning the Post-Impressionist, Cubist, and Abstract Expressionist Movements to a photograph and found it worked successfully on all styles. Finally we experimented with altering the model's loss functions to determine their effect on the resulting image.

1 Introduction and related work

Neural Style Transfer was first introduced by Gatys et al. (1) in 2016. This algorithm leverages the latent spaces that exist in neural network models to identify and extract the content and style layer representations of images. Separating these two allows the model to optimize for an image to match one reference along the content axis and to match another along the style axis. Specifically, by employing a pretrained image recognition neural network to extract features of an image, encoded at different levels in the neural network, we can create a loss function that maximizes similarity in the features themselves (content) or in patterns among the features (style).

The content loss function is simply the mean squared error between the feature representations, since that is what the original network is meant to capture. The style loss function takes the Gram matrix—simply the feature map multiplied by its own transpose—which encode *correlations* between each pair of feature. Gatys et al. show that we can describe the style of an image with these correlations. Importantly, he used features from several layers throughout the network to balance transfer of low-level and high-level features.

In the almost ten years since the Gatys paper, neural style transfer has become a significant field of research; style transfer tools are available online and have had commercial applications in creative fields and marketing. Several improvements have been made to the relatively simple algorithm initially proposed, which struggled somewhat with local information and smaller features (2). Our project consists of three parts: first, replicating the methods described in the Gatys paper; second, testing how well the network can transfer different artistic styles; and third, modifying the model's Gram matrix loss function with a Fourier transform function. We were successful in replicating the model and analyzed its performance on artworks of different styles. We also successfully added spectral differences to the style loss function and found that it transferred slightly different features from the Gram matrix.

2 Datasets

In the original paper, two of the images that are used to test their model are *Starry Night* by Vincent van Gogh(3) and a photograph of Tübingen, Germany taken by Andreas Praefcke in 2003 (4). The image files they used were not provided by them, but we were able to find open-source copies of both files on Wikimedia and Wikipedia respectively so that we could fully their results. Additionally, we chose to use 2 other paintings in different artistic styles as our style image to determine which

style transfers best to the Praefcke photograph. Specifically, we chose to use *Convergence* by Jackson Pollock (5), an abstract piece, and *Still life with Guitar* by Pablo Picasso (6), a Cubist piece, as additional style images.

3 Methods

Like the original paper, we utilized the existing VGG-19 architecture to build our style transfer model. We implemented separate content and style loss functions that were weighed relatively and summed. This allows the model take both content and style into account during loss minimization. We took our losses from the layers the author identified as optimal.

The content loss function is defined as

$$L_{\text{content}}(\vec{p}, \vec{x}, l) = \frac{1}{2} \sum_{i,j} (F_{i,j}^l - P_{i,j}^l)^2$$

where $\vec{a}$ and $\vec{x}$ are the original input image and the image being generated at each iteration of the gradient descent, and P^l and F^l are their respective feature representations in layer l .

The style loss function generates a feature correlation by utilizing a Gram matrix that is defined as $G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l$ where G_{ij}^l is the inner product between the vectorized feature map i and j in layer l . Ultimately the style loss function is

$$L_{\text{style, Gram}}(\vec{a}, \vec{x}) = \sum_{l=0}^L w_l \frac{1}{4N_l^2 M_l^2} \sum_{i,j} (G_{i,j}^l - A_{i,j}^l)^2$$

where w_l is the weight determining the contribution of each layer to the loss, $\vec{a}$ and $\vec{x}$ are the original input image and the image being generated at each iteration of the gradient descent, and A^l and G^l are their respective style representations (or Gram matrices) in layer l .

When we modified the model, we added a term for spectral loss, computed simply as the 2D Fourier transform of over the two feature map dimensions. The style loss function then included an extra term for the average magnitude difference between the two spectra, with a scaled correction term to avoid Fourier reconstruction of the original image:

$$L_{\text{style, FFT}}(\vec{a}, \vec{x}) = \|\text{FFT}(\vec{x}) - \text{FFT}(\vec{a})\|^2 - \kappa L_{\text{content}}(\vec{a}, \vec{x})$$

Since our content loss function returned errors several orders of magnitude greater than the Gram matrix style loss, we added another parameters α and β to control their relative strength. We also added a scaling factor γ to account for the difference in average values between the Gram and spectral losses; by trial and error, we set $\alpha = 2$, $\beta = 2 \times 10^5$, $\gamma = 5 \times 10^{-9}$, $\kappa = 100$. Finally, we allowed ourselves to tune the fraction f of each source in the combined loss. In total:

$$L = \alpha L_{\text{content}} + \beta (f L_{\text{style, Gram}} + \gamma (1 - f) L_{\text{style, FFT}})$$

No specific optimization function for the model was specified in the original paper, but the author later suggested using L-BFGS, so that is the optimizer we chose.



Figure 1: Tübingen, Germany by Andreas Praefcke (2003). Photograph used as the content image for each experiment.

After building our model we ran 3 style transfer experiments using the paper’s original loss functions over 300 iterations, changing the style image while keeping the content image and the weights for the respective loss functions constant. The 3 style images we used were *Starry Night* by Vincent van Gogh (3), *Still life with Guitar* by Pablo Picasso (6), and *Convergence* by Jackson Pollock (5). The content image we used was a photograph of Tübingen, Germany taken by Andreas Praefcke in 2003 (4), which was used in the original paper.

4 Results and experiments



Figure 2: Style Images in top row. Style transfer result in bottom row. From left to right: *Starry Night*. Van Gogh (1889), *Still Life with Guitar*. Picasso (1942), *Convergence*. Pollock (1952).

Figure 2 illustrates the style transfer results using three different painting styles on the same content image. Because style is inherently subjective, we do not use any quantitative metrics to evaluate our output and neither does the original paper. Qualitatively we can see that all three transfer attempts were successful to some degree, with all the outputs maintaining the larger structures from the content image (like the buildings) while also incorporating the colors and textures from each style image. To our eyes it seems as though the transfer attempt where *Convergence* (5) was used as the style image

did the best at maintaining the smaller details from the content image such as the building windows, while the attempt using *Still Life with Guitar*. (6) did the worst at maintaining finer content features. We see that our network does best on larger features—an issue others have pointed out (2).

Finally, focusing on *Starry Night*, we tuned the fraction f of the loss function between the spectral component and the Gram component, with the results shown in Figure 3. We can see that, while the Gram style captured more long-range aspects of the style, like the strong dark borders and the shading levels, it misses some of the smaller texture like the swirls of the *Starry Night*. Conversely, the spectral loss function tends to emphasize apply local textures, making the sky in the spectral loss images seem more dynamic. There is more contrast in the images with more Gram loss; it seems that this function does a better job of mapping specific features of similar parts of the image rather than applying a style like a uniform wash. We think, subjectively, that the Gram function is overall more effective; however adding a small amount of spectral loss can help add some texture, as in the $f = 0.8$ image. It is interesting to explore implementations where the Gram loss is not replaced but instead complemented by other types of loss.

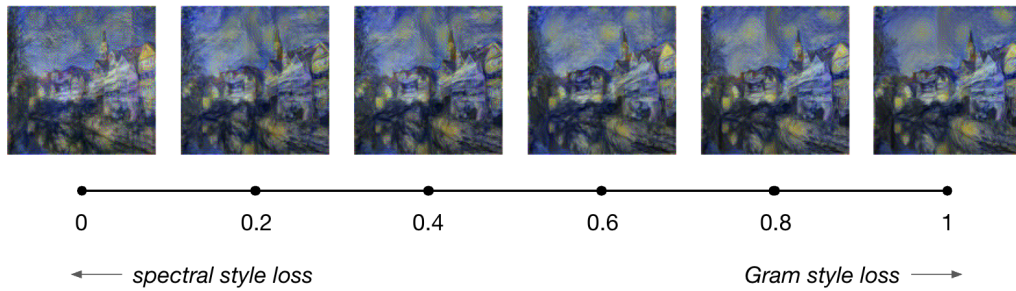


Figure 3: Images generated tuning the amount of loss from Gram matrices and spectral loss.

5 Conclusion and future work

Overall our implementation of Neural Style Transfer was successful. We developed a model that is capable of transferring various artistic styles to a content image while still preserving the content image’s core features. In the future, one might want to test more artistic styles or less geometric content images to determine whether there is a limit to the model’s transfer ability. With more time, we might also modify the spectral loss function to allow a user to tune the level of detail they want to transfer. A Fourier transform offers a simple way to implement low pass and high pass filters, creating a simple means by which users can control whether the network should focus on long-range correlations or local ones.

6 Broader impacts

This style transfer model has the potential to negatively impact the current artistic landscape. While this model does not train on large amounts of art, it does still require the input of two artistic pieces in order to generate our output image. All of the data that we used in our project is open source, but there is nothing stopping someone from feeding in style or content images that were taken without the original artist’s consent to create an output. Additionally, we are not planning on sharing our results beyond this report, but there are serious ethical implications if someone chose to present the output of this model as their own work if they did not produce both the style and content inputs.

7 Code

Our code and the image files we used to train on are located in the following GitHub repository: https://github.com/StephanieFulcar/CSCI152_Final_Project_Neural_Style_Transfer.git. There are two branches in our codebase: main and Fourier. The main branch

contains our Artistic Style Variation experiment while the Fourier branch houses our spectral style loss implementation and tests.

References

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